Michelle Buchi Okonicha

AI Engineer · LLMs, Agents and Evaluation · Google Developer Expert (GDE)
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PROFILE

AI Engineer and Google Developer Expert with hands-on experience shipping LLM products to real users, including RAG pipelines, agentic workflows and multi-modal applications. My current MSc research builds an evaluation pipeline that measures and mitigates goal drift in recursive agents, so I care a lot about tracing, monitoring and testing how agents actually behave in production. Completing an MSc in AI and Data Science at the University of Hull in January 2027 and available to start that month.

TECHNICAL SKILLS

LLMs and Agents LLM applications, RAG, LangChain, agentic workflows (LangGraph), tool calling, prompt engineering, semantic search, MongoDB Vector Search, fine-tuning (LoRA), Hugging Face
Evaluation and MLOps Agent and LLM evaluation, benchmarking, survival analysis for model comparison, MLflow, model monitoring (drift and latency), Docker, Kubernetes, CI/CD
Models and Data PyTorch, TensorFlow, computer vision (YOLO26, YOLOv8, Mask R-CNN), NLP, quantisation, ONNX Runtime
Engineering Python, FastAPI, REST APIs, JavaScript, AWS, GCP, PostgreSQL, MongoDB, Supabase

PROFESSIONAL EXPERIENCE

AI Engineer, Evaluation and Benchmarking | Crossing Party Lines *Oct 2026 - Present · Part-time*

- Building a closed-book LLM benchmark for a US nonprofit from official congressional bill data: 1,583 questions across 300 bills, over six question templates from easy to hard.
- Reviewed and labelled candidate datasets, then designed the question templates with balanced answers and majority-class baselines.
- Built an independent verification step that recomputes every gold answer from the raw XML.
- Built the evaluation harness and abstention-aware scoring, reporting precision, attempt rate and hallucination rate across Llama 3.1 8B, Qwen2.5 7B and DeepSeek.

Machine Learning and AI Engineer | PLATE *Aug 2025 - Jun 2026*

- Built the AI layer for PLATE, a kitchen assistant app live on Google Play, combining real-time computer vision food recognition with prompt-engineered multi-modal LLMs for automated recipe generation.
- Evaluated several multi-modal models against each other for accuracy, cost and latency before choosing the production model.
- Built an offline food recognition system with YOLO26n, trained on around 149,000 images merged from 5 datasets, so recognition works on-device without an internet connection.
- Built async FastAPI model-serving APIs, containerised with Docker, with automated logging to track model drift through continuous user feedback.
- Optimised the deep learning CV pipeline and cut inference latency by 35% for cross-platform mobile users.

AI Software Engineer | Aricah Technologies *Jan 2024 - Dec 2025*

- Engineered and shipped 3 production AI applications for clients across edtech and food logistics (Velarius, Frootaz and Belish), available on web and Google Play Store.
- Built Velarius, an LLM-powered learning platform for secondary school students, using LangChain to orchestrate RAG over OpenAI and Gemini models, with secure document indexing and context-grounded tests. 50+ downloads.
- Architected RAG pipelines using MongoDB Vector Search, improving context retrieval precision by 22%.
- Built FastAPI services to serve LLM features, handling context-window limits across user sessions.

Software Engineer | Landmark Africa *Jan 2022 - Jan 2024*

- Maintained applications serving 200,000+ monthly customers and contributing circa ₦2bn in revenue.
- Built the Hospitality Institute student portal and admin panel, reducing administrative bottlenecks through automated workflows.
- Developed the Upsidedown House app, the first interactive experience of its kind in West Africa, logging data for 100,000+ event visitors.

Software Support Engineer | Zanibal LLC *Mar 2021 - Nov 2021*

- Improved bug resolution cycles by 25% through SQL reporting pipelines and batch script automation for enterprise wealth management clients.
- Promoted within 3 months, conducting penetration testing on the fintech platform and onboarding enterprise clients.

OTHER KEY PROJECTS

Goal Drift in Recursive Agentic Workflows on Edge-Quantised SLMs *MSc Research · Ongoing*

- Building an evaluation pipeline that measures and mitigates goal drift in stateless small language model agents.
- Implemented a LangGraph auditor that monitors agent behaviour across recursive steps, benchmarking IBM Granite 4.2 models (3B and 8B) on Apple hardware and HPC.
- Modelling time-to-drift with survival analysis (Kaplan-Meier, Cox) to compare models and mitigation strategies.

AfroLearner *In Development · Open-Source Lead*

- NLP language learning platform for low-resource African languages (Wolof) built with Python, FastAPI and the Waxal dataset, with custom tokenisers and local inference via Meta's MMS framework.

EDUCATION

MSc Artificial Intelligence and Data Science · University of Hull, UK *Expected Jan 2027*

Advanced machine learning, neural networks, computer vision, big data and research methods in AI.

BSc Applied Artificial Intelligence · IU University of Applied Sciences

Natural language processing, linear algebra, Python programming, reinforcement learning, computer vision, statistics.

LEADERSHIP AND COMMUNITY

Google Developer Expert (GDE)

- Recognised by Google for technical expertise and developer community leadership. Delivered 60+ technical talks, webinars and bootcamps and written almost 40 technical articles.

Partnerships Lead, AI Society · University of Hull

- Build industry partnerships that bring AI workshops, hackathons and technical resources to students.

Girls Like Me Dream Initiative

- Outreach and AI training programme that has reached over 640 girls.

AWARDS AND RECOGNITION

Full Circle Award, Google Showcase *2026*

First Google Developer Expert to emerge from the Women Developer Academy in Sub-Saharan Africa, now mentoring others through the GDE Academy.

DAIM Hackathon, 1st Place, AI Racer (University of Hull) *2026*

Winning team in an autonomous racing challenge combining a PPO reinforcement learning agent with a NEAT-style genetic algorithm.

Featured in The Geek Circle · Read the feature *2026*

Profiled on my path into AI and machine learning and my work encouraging girls into technology.

Women Techmakers High Impact Award *2023*

First recipient in Sub-Saharan Africa, for outstanding contributions to the technical community.